Indian Institute of Technology Jodhpur
Department of Mathematics
Mathematics II MAL1020
Problem Sheet II

AY 25-26 Semester II

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1. Let

$$v_1 = (1, 2, 1, -1), \quad v_2 = (2, 1, -1, 1), \quad v_3 = (3, 3, 0, 0), \quad v_4 = (1, 5, 3, -3)$$

be vectors in $\mathbb{R}^4$.

   (a) Determine whether v_4 can be expressed as a linear combination of v_1, v_2, and v_3.

   (b) If possible, find scalars α, β, γ such that

$$v_4 = \alpha v_1 + \beta v_2 + \gamma v_3.$$

   (c) Determine whether the representation is unique.

   (d) Find all linear combinations of v_1, v_2, v_3 that yield the zero vector.

2. Let

$$S = \{(1, 1, 1, 0),\ (2, 1, 0, 1),\ (3, 2, 1, 1),\ (1, 0, -1, 1)\} \subset \mathbb{R}^4.$$

   (a) Test whether the set S is linearly independent.

   (b) Extract a basis for $\text{span}(S)$.

   (c) Determine $\dim(\text{span}(S))$.

   (d) Express the vector $(4, 2, 0, 2)$ as a linear combination of the basis vectors obtained.

3. Let

$$U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\} \quad \text{and} \quad W = \text{span}\{(1, 1, 1)\}.$$

   (a) Prove that $\mathbb{R}^3 = U \oplus W$.

   (b) For the vector $v = (2, -1, 3)$, find unique vectors $u \in U$ and $w \in W$ such that

$$v = u + w.$$

   (c) Verify explicitly that $U \cap W = \{0\}$.

4. Let

$$U = \text{Col}\left(\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}\right) \subset \mathbb{R}^3.$$

   (a) Find a subspace $W \subset \mathbb{R}^3$ such that

$$\mathbb{R}^3 = U \oplus W.$$

(b) Verify that $U + W = \mathbb{R}^3$ and $U \cap W = \{0\}$.

(c) Decompose the vector $(3, 2, 1)$ uniquely as

$$(3, 2, 1) = u + w, \quad u \in U,\ w \in W.$$

5. Let

$$B = \{(1, 1, 0),\ (0, 1, 1),\ (1, 0, 1)\}$$

be an ordered basis of $\mathbb{R}^3$.

(a) Find the change-of-coordinates matrix $P_{B \to E}$ from basis B to the standard basis.

(b) Compute the inverse matrix $P_{E \to B}$.

(c) Find the coordinate vector $[v]_B$ of $v = (2, 3, 1)$.

6. Determine whether the following set of vectors spans $\mathbb{R}^2$:

(i) $\{(1, -1), (2, -2), (2, 3)\}$.

(ii) $\{(6, -2), (-2, 2/3), (3, -1)\}$.

7. Show that the set of vectors

$$\{(1, 2, 3), (3, 4, 5), (4, 5, 6)\}$$

does not span $\mathbb{R}^3$, but that it does span the subspace of $\mathbb{R}^3$ consisting of all vectors lying in a plane (find the equation of the plane).

8. Solve the following problems:

(i) Consider the vectors

$$A_1 = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 1 \\ -2 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}$$

in $\mathscr{M}_2(\mathbb{R})$. Determine $\operatorname{span}\{A_1, A_2, A_3\}$.

(ii) Consider the vectors

$$A_1 = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}, \quad A_2 = \begin{pmatrix} -2 & 1 \\ 1 & -1 \end{pmatrix}$$

in $\mathscr{M}_2(\mathbb{R})$. Find $\operatorname{span}\{A_1, A_2\}$, and determine whether or not $\begin{pmatrix} 3 & 1 \\ -2 & 4 \end{pmatrix}$ lies in the subspace.

9. Determine whether the following vectors are linearly dependent/independent:

(i) $\mathbf{v}_1 = (1, 4, 1, 7), \mathbf{v}_2 = (3, -5, 2, 3), \mathbf{v}_3 = (2, -1, 6, 9), \mathbf{v}_4 = (-2, 3, 1, 6)$ in $\mathbb{R}^4$.

(ii) The vectors $p_1(x) = a + bx$ and $p_2(x) = c + dx$ (find conditions on a, b, c, and d).

10. Check whether the following set of vectors spans $\mathscr{P}_2(\mathbb{R})$:

$$\{2 + x^2, 4 - 2x + 3x^2, 1 + x\}.$$

11. Let $p_1(x) = 1 + x$, $p_2(x) = x(x - 1)$, and $p_3(x) = 1 + 2x^2$. Determine the coordinates of an arbitrary polynomial $p(x) = a_0 + a_1 x + a_2 x^2$ relative to the ordered basis $\{p_1, p_2, p_3\}$.